Appendix: Conservative Event-Level Timing Audit

Appendix A
Conservative Event-Level Timing Audit

Each logged REFIT row provides a common timestamp but not the physical acquisition time of every individual meter reading. Consequently, the recorded data cannot identify an exact, universal IAM-aggregate delay. A single nearby aggregate step is also insufficient evidence for a shift, since an unmonitored load can produce that step. We therefore describe the following procedure only as a conservative, diagnostic event-level audit; it is not part of the primary training or evaluation pipeline and does not claim physical synchronization.

Let x_t and $P_{t,i}$ denote the recorded aggregate and appliance powers, respectively, and define $\Delta x_t = x_t - x_{t-1}$. Consider a complete on period $[s, e)$ of appliance i, with a rising edge $a_i^{\uparrow} = P_{s,i} - P_{s-1,i}$ and a falling edge $a_i^{\downarrow} = P_{e-1,i} - P_{e,i}$. Only events whose two appliance edges are at least $\max(40, 0.3\tau_i)$ W are considered, where τ_i is the appliance on threshold. We examine lags $\ell \in \{-2, -1, 0, 1, 2\}$, corresponding to at most ± 16 s. A lag is admissible only if both aggregate edges have the expected direction and sufficient magnitude:

$$\begin{aligned} \Delta x_{s+\ell} &\geq \max(20, 0.25 a_i^{\uparrow}), \\ -\Delta x_{e+\ell} &\geq \max(20, 0.25 a_i^{\downarrow}). \end{aligned} \tag{1}$$

For every admissible lag, the two-edge discrepancy is

$$C_i(\ell) = \frac{1}{2}\left(\frac{|\Delta x_{s+\ell} - a_i^{\uparrow}|}{\max(a_i^{\uparrow}, 40)} + \frac{|-\Delta x_{e+\ell} - a_i^{\downarrow}|}{\max(a_i^{\downarrow}, 40)} \right). \tag{2}$$

The selected lag is $\ell^* = \arg\min_\ell C_i(\ell)$, with zero lag preferred only when scores are tied. We accept it only when $C_i(\ell^*) \leq 0.75$, its margin over the next-best admissible lag is at least 0.20, neither event boundary lies near a ≥ 40 W transition of another monitored target, and the shifted event would not overlap an existing on period. Events failing any condition are left unchanged.

For an accepted non-zero lag, the aggregate is immutable. Let b_i be the median measured power in the nearby off samples and let $q_{t,i} = \max(P_{t,i} - b_i, 0)$ within $[s, e)$. The procedure moves the excess event profile as a whole:

$$\widetilde{P}_{t,i} = P_{t,i} - q_{t,i}\mathbf{1}_{[s,e)}(t) + q_{t-\ell^*,i}\mathbf{1}_{[s+\ell^*,e+\ell^*)}(t), \qquad \widetilde{x}_t = x_t. \tag{3}$$

Thus $\sum_t \widetilde{P}_{t,i} = \sum_t P_{t,i}$ for every shifted event. The associated binary on period is shifted by the same lag. This preserves the recorded input and the appliance watt-sample energy; it does *not* impose $x_t \geq \sum_i P_{t,i}$, because the measured target set is incomplete and the recorded aggregate must not be reconstructed using its labels.

The audit is a sensitivity analysis, not part of the retained training pipeline. In our cross-house ablation it slightly changed local timing errors but did not consistently improve F1 or AP across the two held-out datasets. The primary results therefore use the unshifted, mean-resampled recordings. This distinction avoids treating a plausible timing hypothesis as ground truth.

Algorithm 1 Conservative two-edge event timing audit.

Require: 8-s aggregate x_t, appliance powers $P_{t,i}$, states $z_{t,i}$, thresholds τ_i, and $L = 2$ samples.
Ensure: Audited appliance series $\widetilde{P}_{t,i}$ and states $\widetilde{z}_{t,i}$; aggregate x_t is unchanged.

```
Set P̃_{t,i} ← P_{t,i} and z̃_{t,i} ← z_{t,i}.
for each target appliance i ∈ {1, ..., M} do
    for each complete on period [s, e) in z_{t,i} do
        Reject the event if an endpoint is unavailable, either appliance edge is below max(40, 0.3τ_i), or another monitored target has a transition of at least 40 W within L samples of either boundary.
        Form the admissible lag set using (1); discard candidates whose two relative edge errors exceed one.
        if C_i(ℓ*) ≤ 0.75, the score margin is at least 0.20, and the shifted interval does not overlap another ON interval then
            Move the excess profile and the complete state interval by ℓ* using (3).
        end if
    end for
end for
Verify Σ_t P̃_{t,i} = Σ_t P_{t,i} for every appliance and x_t is bitwise unchanged.
return x_t, P̃_{t,i}, and z̃_{t,i}.
```

Appendix B
Loss-Function Ablation and Hyperparameter Robustness

This appendix documents the loss-function study used to finalize MART-NILM. The retained objective removes relative energy loss (LRE), retaining on-state mean squared error (ON-MSE), off-state mean squared error (OFF-MSE), on-state masked temporal-difference mean squared error (Delta-MSE), a false-positive (FP) state penalty, and binary cross-entropy (BCE) state supervision. The selected coefficients are $w_{\text{on}} = 1.0$, $w_{\text{off}} = 0.5$, $w_{\Delta} = 0.15$, $w_{\text{FP}} = 1.0$, and $\lambda_{\text{state}} = 0.8$; the BCE positive-weight cap is 12 and the task-balance clipping limit is 3.

A. Objective Before and After LRE Removal

Let Q_i, S_i, and r_i denote the base power loss, state loss, and detached appliance-wise balance ratio for appliance i,

respectively. With $\mathrm{sg}(\cdot)$ denoting stop-gradient, define

$$\begin{aligned} Q_i &= \mathrm{MSE}_i + w_{\mathrm{on}}\,\mathrm{ON\text{-}MSE}_i + w_{\mathrm{off}}\,\mathrm{OFF\text{-}MSE}_i \\ &\quad + w_{\Delta}\,\mathrm{Delta\text{-}MSE}_i, \\ S_i &= \mathrm{BCE}_i + w_{\mathrm{FP}}\,\mathrm{FP}_i\,. \end{aligned} \tag{4}$$

The original objective used for the frozen Table III reference was

$$\begin{aligned} \mathcal{L}_{\text{with-LRE}} &= \sum_{i=1}^{A} (Q_i + 0.25\,\mathrm{LRE}_i) + 0.8 \sum_{i=1}^{A} S_i\,\mathrm{sg}\big(r_i^{\mathrm{old}}\big)\,, \\ r_i^{\mathrm{old}} &= \mathrm{clip}\left(\frac{Q_i + 0.25\,\mathrm{LRE}_i}{S_i}, \frac{r^{\mathrm{old}}}{3}, 3r^{\mathrm{old}}\right), \\ r^{\mathrm{old}} &= \frac{\sum_i (Q_i + 0.25\,\mathrm{LRE}_i)}{\sum_i S_i}, \\ \mathrm{LRE}_i &= \frac{\sum_{t=1}^{T} |\widehat{P}_i(t) - P_i(t)|}{\sum_{t=1}^{T} P_i(t) + T P_0}, \qquad P_0 = 10\ \mathrm{W}. \end{aligned} \tag{5}$$

Here, FP_i is the mean squared state probability on true-OFF samples, and Delta-MSE is evaluated only on temporal pairs for which at least one sample is truly ON. The terms MSE, ON-MSE, OFF-MSE, and Delta-MSE are evaluated in the standardized power space, whereas LRE is evaluated after conversion to watts.

The final retained objective is identical except that the relative-energy term is removed:

$$\begin{aligned} \mathcal{L}_{\mathrm{final}} &= \sum_{i=1}^{A} Q_i + 0.8 \sum_{i=1}^{A} S_i\,\mathrm{sg}\big(r_i^{\mathrm{final}}\big)\,, \\ r_i^{\mathrm{final}} &= \mathrm{clip}\left(\frac{Q_i}{S_i}, \frac{r^{\mathrm{final}}}{3}, 3r^{\mathrm{final}}\right), \\ r^{\mathrm{final}} &= \frac{\sum_i Q_i}{\sum_i S_i}, \\ Q_i &= \mathrm{MSE}_i + 1.0\,\mathrm{ON\text{-}MSE}_i + 0.5\,\mathrm{OFF\text{-}MSE}_i \\ &\quad + 0.15\,\mathrm{Delta\text{-}MSE}_i, \\ S_i &= \mathrm{BCE}_i + 1.0\,\mathrm{FP}_i\,. \end{aligned} \tag{6}$$

Thus $w_E = 0$ and neither LRE nor its denominator floor P_0 remains in the final model. The ON-class BCE weight is still capped at 12, and r_i retains the same clipping limit $c = 3$.

All candidates retain the same data split, architecture, optimization schedule, post-processing, and evaluation pipeline. A term is removable or a replacement retainable only when it maintains or improves power-estimation and state-detection metrics in both test houses. Confirmed comparisons use three seeds (2026, 2027, and 2028); one-seed screens are used only to reject clearly weak neighbouring values.

B. OFF-MSE Waveform and Background-Noise Audit

Tables II and V quantify the OFF-MSE trade-off. Figure 3 provides a compact, matched-event audit: for each appliance, the reference and no-OFF-MSE predictions use the same saved REFIT H20 event and seed. Blue is target power and dashed red is predicted power. The complete background-range panels remain in the experiment archive; they are intentionally not reproduced here because the numerical OFF-MAE, AP, and F1 tables are the primary evidence.

C. Complete Seed-Level Test Results

For auditability, Tables IV-VII report every completed Table III test result underlying the aggregate tables and response curves above. Each row is the "overall" row in the saved metrics file; it is an appliance-averaged result, rather than a cherry-picked appliance result. The retained no-LRE objective and the three confirmed term removals use seeds 2026, 2027, and 2028. The remaining coefficient alternatives and the no-ON-MSE removal are one-seed exploratory screens (seed 2026), so they are reported as such and are not evidence of a statistically significant ranking.

State metrics alone are insufficient: removing OFF-MSE raises REFIT OFF-MAE from 7.55 to 8.21 W despite similar F1 (0.717 to 0.715), and removing Delta-MSE raises UK-DALE ON-MAE from 175.30 to 202.03 W despite AP changing only from 0.926 to 0.924. The evidence therefore supports the compact retained objective within the tested ranges, not a claim of formal global hyperparameter optimality.

Appendix C
Controlled Training-Strategy Comparisons and Concurrent-Activation Audit

The controlled comparisons use the final no-LRE objective, the fixed five-week household split, and seeds 2026, 2027, and 2028. Each intervention changes one mechanism: (i) random recomposition disables focal-event selection and residual-power stratification while retaining the 50% recomposition probability; (ii) global balancing replaces local ratios with the batch-global power-to-state loss ratio; (iii) unbounded appliance-wise balancing removes the global bounds from local ratios; and (iv) the relation-attention ablation disables only cross-appliance attention. All remaining losses, coefficients, architecture settings, and decoding rules are held fixed. Checkpoints and state thresholds use validation data only; held-out test results are report-only.

For the concurrency audit, samples are grouped by the number of recorded active target appliances, $c_t = \sum_i z_{t,i}$, into 0, 1, 2, 3, and 4–5 active targets. Paired model comparisons require identical sample alignment and recorded targets. Each group reports its sample count, power MAE, ON MAE, macro-F1, and the exact-match proportion $|\mathcal{G}|^{-1} \sum_{t \in \mathcal{G}} \mathbb{I}[\hat{\mathbf{z}}_t = \mathbf{z}_t]$. Per-appliance F1 is undefined when there are no recorded or predicted positives; ON MAE is undefined when there are no recorded ON samples. Such entries are excluded from the group macro average and their available-class counts are reported. All-OFF samples are reported separately to expose their influence on exact match. These analyses evaluate frozen models and do not select an architecture or tune a decision threshold.

TABLE I

THREE-SEED COMPARISON BEFORE AND AFTER REMOVAL OF LRE. MEAN ± STANDARD DEVIATION. **SELECTED** IDENTIFIES THE FINAL OBJECTIVE. REMOVING LRE ELIMINATES THE DENOMINATOR FLOOR $P_0 = 10$ W WHILE PRESERVING PERFORMANCE.

Test house	Objective	MAE (W)	ON-MAE (W)	SAE (%)	AP	F1	Decision
UK-DALE H2	With LRE	7.451 ± 0.091	176.669 ± 7.675	5.275 ± 0.100	0.926 ± 0.008	0.876 ± 0.010	reject
UK-DALE H2	**Reference: no LRE**	7.296 ± 0.173	175.296 ± 6.924	5.059 ± 0.364	0.926 ± 0.010	0.880 ± 0.003	**Selected**
REFIT H20	With LRE	10.798 ± 0.355	309.207 ± 4.577	5.893 ± 0.390	0.762 ± 0.018	0.716 ± 0.016	reject
REFIT H20	**Reference: no LRE**	10.931 ± 0.174	309.988 ± 3.540	5.959 ± 0.233	0.761 ± 0.016	0.717 ± 0.003	**Selected**

TABLE II

LOSS-TERM REMOVALS RELATIVE TO THE NO-LRE REFERENCE. VALUES ARE THREE-SEED MEANS UNLESS MARKED *screen* ($n = 1$). THE BOLD REFERENCE ROWS ARE RETAINED; ALL REMOVALS ARE REJECTED UNDER THE CROSS-HOUSE DECISION RULE. OFF-MAE AUDITS BACKGROUND-POWER NOISE.

Test house	Variant	n	MAE (W)	ON-MAE (W)	OFF-MAE (W)	SAE (%)	AP / F1	Decision
UK-DALE H2	**Reference**	3	7.296	175.296	4.830	5.059	.926 / .880	**Retain**
	No ON-MSE (*screen*)	1	8.217	310.657	4.115	6.136	.920 / .874	reject
	No OFF-MSE	3	7.161	174.972	4.981	4.801	.925 / .877	reject
	No Delta-MSE	3	7.472	202.025	4.796	5.251	.924 / .865	reject
	No FP penalty	3	7.257	177.210	4.850	5.024	.919 / .874	reject
REFIT H20	**Reference**	3	10.931	309.988	7.550	5.959	.761 / .717	**Retain**
	No ON-MSE (*screen*)	1	11.238	432.794	5.607	6.702	.723 / .688	reject
	No OFF-MSE	3	11.184	307.688	8.212	6.200	.755 / .715	reject
	No Delta-MSE	3	10.716	316.605	7.294	5.725	.757 / .725	reject
	No FP penalty	3	10.858	304.509	7.572	5.944	.760 / .716	reject

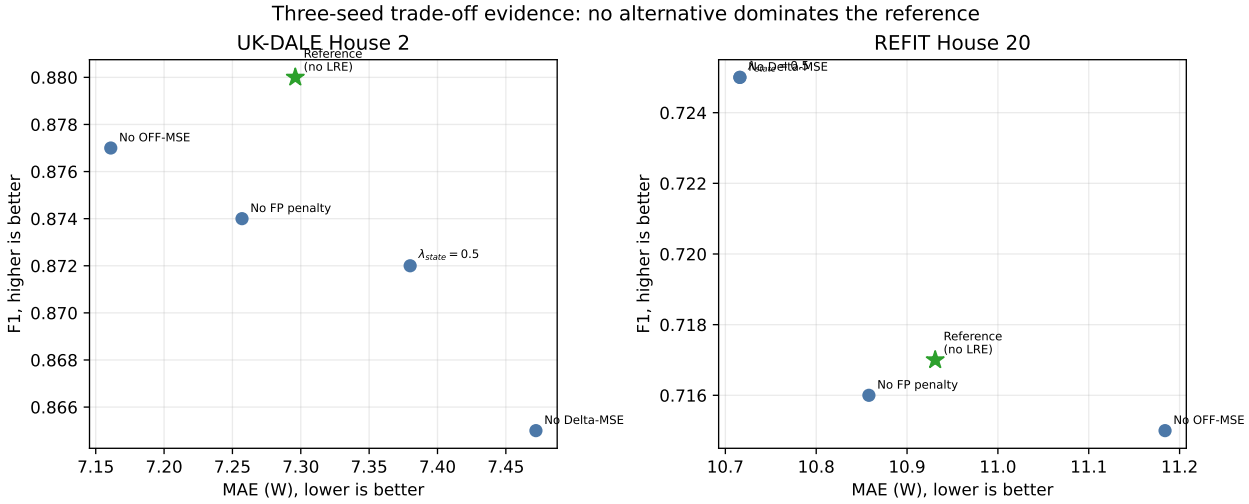


Fig. 1. Three-seed Pareto view of confirmed loss-term and weight alternatives. The no-LRE reference is retained because no alternative achieves lower MAE and higher F1 in both test houses.

TABLE III

COMPACT ONE-SEED SCREEN DATA. EACH SEQUENCE FOLLOWS INCREASING TESTED VALUES AND GIVES THE ARITHMETIC MEAN ACROSS UK-DALE H2 AND REFIT H20. BOLD IDENTIFIES THE RETAINED SETTING, NOT THE NUMERICAL MINIMUM OF AN ISOLATED METRIC; NO CANDIDATE GIVES A STRICT JOINT POWER-STATE IMPROVEMENT.

Parameter	Tested values	Two-house mean MAE (W)	Two-house mean F1	Decision
w_{on}	.5, **1.0**, 2.0	9.257, 9.177, 9.351	.796, .793, .788	**Retain 1.0**
w_{off}	.25, **.5**, 1.0	9.208, 9.177, 9.201	.792, .793, .786	**Retain .5**
w_{Δ}	.05, **.15**, .30	9.230, 9.177, 9.137	.785, .793, .796	**Retain .15**
w_{FP}	.5, **1.0**, 2.0	9.004, 9.177, 9.141	.794, .793, .797	**Retain 1.0**
λ_{state}	.5, **.8**, 1.0	8.942, 9.177, 9.421	.799, .793, .783	**Retain .8**
BCE cap	6, **12**, 24	9.185, 9.177, 9.112	.791, .793, .791	**Retain 12**
Balance limit	2, **3**, 5	9.048, 9.177, 9.243	.801, .793, .793	**Retain 3**

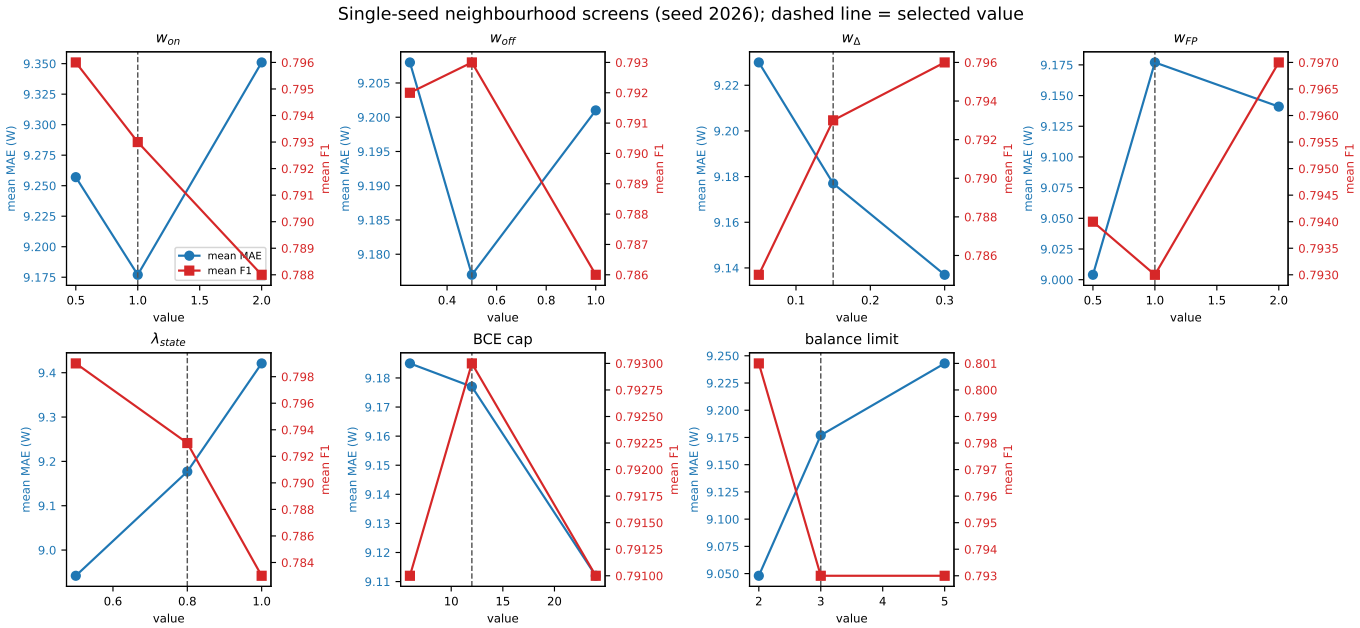
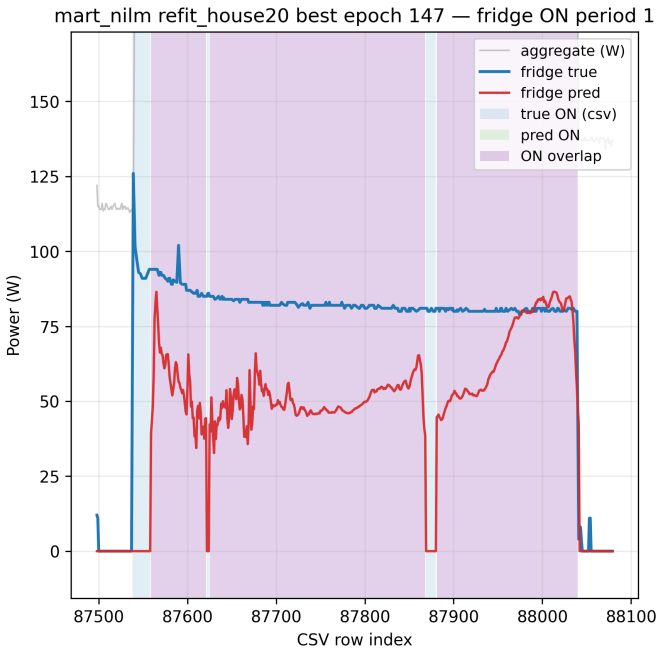


Fig. 2. Single-seed local hyperparameter screens. Blue: two-house mean MAE (lower is better). Red: two-house mean F1 (higher is better). Dashed lines mark the retained values.

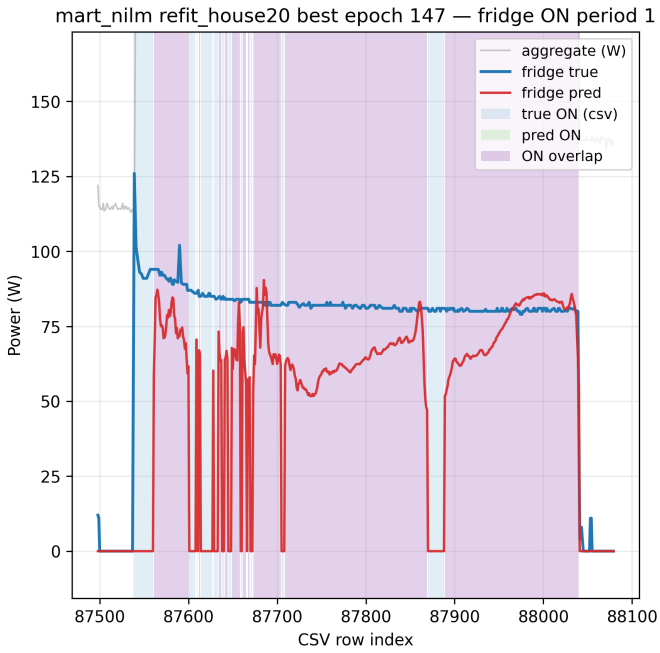
TABLE IV

COMPLETE THREE-SEED TABLE III RESULTS FOR THE RELATIVE-ENERGY-LOSS (LRE) COMPARISON. EACH SEED WAS EVALUATED AT THE VALIDATION-SELECTED CHECKPOINT. MAE, ON-MAE, AND OFF-MAE ARE IN W; SAE IS IN %.

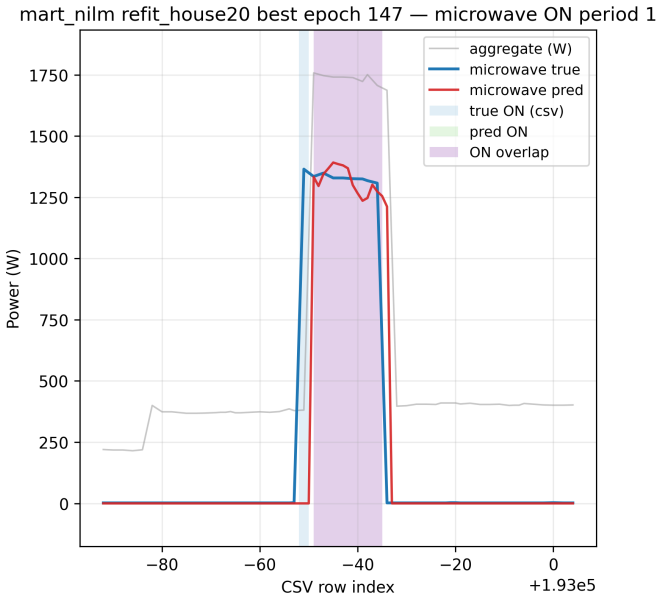
Test house	Objective / seed	MAE	ON-MAE	OFF-MAE	SAE	AP	F1
UK-DALE H2	With LRE / 2026	7.407	170.046	4.622	5.287	.935	.879
	With LRE / 2027	7.390	185.080	4.725	5.169	.920	.884
	With LRE / 2028	7.555	174.880	4.831	5.368	.924	.865
	No LRE / 2026	**7.338**	**167.554**	4.772	5.224	**.938**	.877
	No LRE / 2027	**7.106**	**177.439**	4.851	**4.641**	.921	**.883**
	No LRE / 2028	**7.445**	180.896	**4.724**	**5.311**	.920	**.880**
REFIT H20	With LRE / 2026	11.133	310.321	7.513	6.246	.742	.697
	With LRE / 2027	10.836	313.124	7.412	5.960	.777	.722
	With LRE / 2028	**10.426**	304.176	6.723	**5.474**	.768	.728
	No LRE / 2026	11.002	309.573	7.745	6.064	.746	.716
	No LRE / 2027	11.058	**306.674**	7.965	6.120	**.777**	.720
	No LRE / 2028	10.732	313.718	6.985	5.692	.760	.714



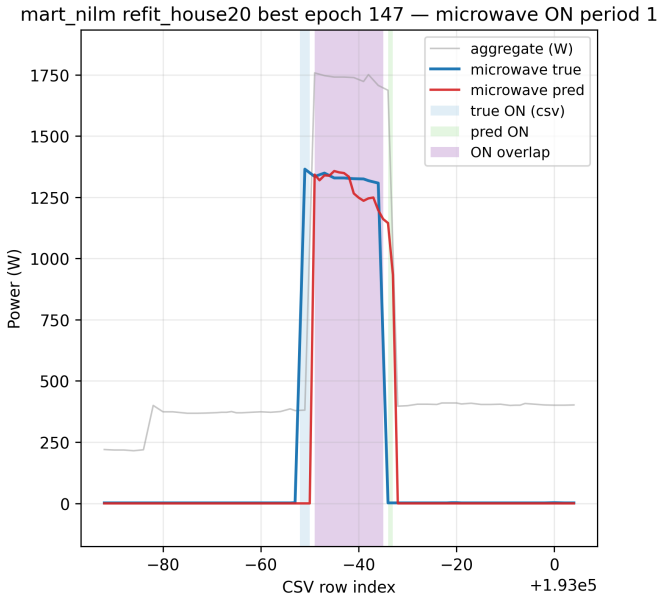
(a) Fridge: retained reference



(b) Fridge: no OFF-MSE



(c) Microwave: retained reference



(d) Microwave: no OFF-MSE

Fig. 3. Compact matched-event waveform audit for seed 2026 on REFIT H20. Each column compares the same saved event under the retained no-LRE reference and no-OFF-MSE removal.

TABLE V

COMPLETE SEED-LEVEL TEST OUTCOMES FOR LOSS-TERM REMOVALS. ALL ROWS USE NO LRE. NO ON-MSE IS A ONE-SEED REJECTION SCREEN; THE OTHER REMOVALS ARE THREE-SEED COMPARISONS.

Test house	Variant / seed	MAE	ON-MAE	OFF-MAE	SAE	AP	F1
UK-DALE H2	No ON-MSE / 2026	8.217	310.657	4.115	6.136	.920	.874
	No OFF-MSE / 2026	7.188	165.192	4.901	5.080	.940	.877
	No OFF-MSE / 2027	7.108	184.424	5.036	4.519	.918	.872
	No OFF-MSE / 2028	7.187	175.301	5.005	4.803	.918	.882
	No Delta-MSE / 2026	7.448	169.168	4.772	5.326	.936	.875
	No Delta-MSE / 2027	7.480	224.361	4.813	5.211	.916	.858
	No Delta-MSE / 2028	7.487	212.547	4.803	5.217	.919	.862
	No FP penalty / 2026	7.126	165.983	4.760	4.954	.932	.879
	No FP penalty / 2027	7.333	200.302	4.853	5.046	.909	.873
	No FP penalty / 2028	7.313	165.344	4.938	5.073	.916	.869
REFIT H20	No ON-MSE / 2026	11.238	432.794	5.607	6.702	.723	.688
	No OFF-MSE / 2026	11.146	303.052	7.904	6.178	.756	.713
	No OFF-MSE / 2027	11.242	313.026	8.641	6.274	.760	.724
	No OFF-MSE / 2028	11.163	306.986	8.091	6.148	.749	.709
	No Delta-MSE / 2026	10.724	301.089	7.419	5.751	.758	.721
	No Delta-MSE / 2027	10.635	328.301	7.072	5.645	.775	.740
	No Delta-MSE / 2028	10.790	320.424	7.391	5.780	.739	.715
	No FP penalty / 2026	10.965	308.647	7.770	6.196	.748	.696
	No FP penalty / 2027	10.938	303.071	7.571	5.881	.769	.730
	No FP penalty / 2028	10.670	301.809	7.374	5.756	.762	.722

TABLE VI

COMPLETE SEED-2026 TABLE III SCREEN RESULTS FOR UK-DALE H2. "SETTING" GIVES THE TESTED VALUE; THE REFERENCE IS THE NO-LRE OBJECTIVE. ALL NON-REFERENCE ROWS ARE EXPLORATORY ONE-SEED SCREENS AND WERE REJECTED.

Screen	Setting	MAE	ON-MAE	OFF-MAE	SAE	AP	F1
Reference	**retained**	**7.338**	**167.554**	4.772	5.224	.938	.877
w_{on}	.5	7.480	180.087	4.611	5.406	.940	.881
w_{on}	2.0	7.256	163.440	5.009	5.029	.938	.874
w_{off}	.25	7.221	165.462	4.881	5.086	.933	.872
w_{off}	1.0	7.555	184.348	4.590	5.621	.937	.879
w_{Δ}	.05	7.273	171.901	4.821	5.078	.925	.870
w_{Δ}	.30	7.332	172.906	4.778	5.214	.940	.874
w_{FP}	.5	7.365	173.195	4.763	5.298	.937	.870
w_{FP}	2.0	7.262	191.412	4.806	5.079	.938	.879
λ_{state}	.5	7.159	177.347	4.896	4.941	.936	.877
λ_{state}	1.0	7.124	196.239	4.757	4.880	.943	.890
BCE cap	6	7.240	172.077	4.669	5.189	.934	.883
BCE cap	24	7.264	162.198	4.821	5.088	.933	.870
Balance limit	2	7.269	168.540	4.772	5.153	.938	.883
Balance limit	5	7.391	169.294	4.821	5.237	.935	.873

TABLE VII

COMPLETE SEED-2026 TABLE III SCREEN RESULTS FOR REFIT H20. THE ROWS CORRESPOND EXACTLY TO TABLE VI; ALL NON-REFERENCE ROWS ARE EXPLORATORY ONE-SEED SCREENS AND WERE REJECTED.

Screen	Setting	MAE	ON-MAE	OFF-MAE	SAE	AP	F1
Reference	**retained**	**11.002**	**309.573**	7.745	6.064	.746	.716
w_{on}	.5	11.034	307.730	7.348	6.264	.747	.711
w_{on}	2.0	11.445	299.844	8.551	6.515	.745	.702
w_{off}	.25	11.195	299.545	8.082	6.160	.754	.711
w_{off}	1.0	10.846	308.349	7.047	6.063	.740	.693
w_{Δ}	.05	11.186	313.238	7.829	6.377	.747	.699
w_{Δ}	.30	10.942	317.410	7.727	5.911	.744	.717
w_{FP}	.5	10.643	320.823	7.335	5.680	.748	.717
w_{FP}	2.0	11.019	318.870	7.756	6.123	.750	.715
λ_{state}	.5	10.724	311.233	7.464	5.786	.766	.720
λ_{state}	1.0	11.717	319.303	8.066	6.631	.731	.675
BCE cap	6	11.130	314.440	7.449	6.195	.734	.698
BCE cap	24	10.959	307.008	7.585	5.907	.751	.711
Balance limit	2	10.826	304.409	7.623	5.952	.751	.718
Balance limit	5	11.094	301.253	7.715	6.036	.743	.712